

SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

Faculty of Computing — Department of Software Engineering

Research Project — Periodic Log Entry Form

Project R26-SE-009 | Agent 2 | Ravindu Bandara Subasinha (IT22098450) | Reporting period 23.03.2026 – 25.08.2026

1. PROJECT AND STUDENT DETAILS

|                          |                                                                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------------------|
| Project ID               | R26-SE-009                                                                                                     |
| Research Topic           | Self-Optimized LLM Multi-Agent System for Software Engineering                                                 |
| Supervisor               | Prof. Nuwan Kodagoda                                                                                           |
| Co-Supervisor            | Mr. Eishan Weerasinghe                                                                                         |
| Student Name             | Ravindu Bandara Subasinha                                                                                      |
| Student Registration No. | IT22098450                                                                                                     |
| Individual Component     | Agent 2 — Code Developer (AI Web Developer) and the Studio user interface                                      |
| Reporting Period         | 23 March 2026 to 25 August 2026                                                                                |
| Phase Covered            | Requirement, implementation and integration phase, representing approximately 65% of the planned project work. |

2. SUMMARY OF RP WORK CARRIED OUT DURING THE REPORTING PERIOD

During this reporting period Agent 2 was developed as the AI Web Developer responsible for turning structured requirements into working software, together with the shared Studio interface through which the whole system is operated. The early part of the period established the generation pipeline: planning, component and page synthesis, packaging, progress logging, sandbox preview and repair support. The middle of the period covered the builder front end and dashboard, followed by a substantial architecture rebuild in which multi-stage architect and planner agents, a component registry with versioning and edit plans, centralised error handling, route resolution with self-healing element patches and a consolidated regression suite were introduced. The later part of the period was spent on model routing, workspace and source-guidance tooling, module splitting for maintainability, and the integration work that brought the SRS, QA and deployment components together with the builder behind one Studio interface and one backend runtime.

### 3. DATE-WISE RP WORK (49 ENTRIES, VERIFIED MONTHLY)

| MARCH 2026                           |            |                                                                                                                                                                   | 5 ENTRIES                 |
|--------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| NO.                                  | DATE       | RP WORK CARRIED OUT                                                                                                                                               |                           |
| 1                                    | 23.03.2026 | Defined the scope of Agent 2 as the AI Web Developer responsible for turning approved requirements into practical software artefacts for web systems.             |                           |
| 2                                    | 24.03.2026 | Reviewed GitHub and open-source project patterns to identify reusable approaches for backend structure, frontend composition, prompt design and repair workflows. |                           |
| 3                                    | 26.03.2026 | Designed the Agent 2 pipeline for converting structured SRS input into component, page, service and project-level outputs.                                        |                           |
| 4                                    | 28.03.2026 | Planned generation support for reusable UI components, page templates, style variants and packaging logic for produced files.                                     |                           |
| 5                                    | 31.03.2026 | Added static planning for build counts, artefact manifests and logged progress so generation runs could be tracked and reviewed.                                  |                           |
| MONTHLY VERIFICATION — MARCH 2026    |            |                                                                                                                                                                   |                           |
| SUPERVISOR'S COMMENTS FOR MARCH 2026 |            | SUPERVISOR'S SIGNATURE                                                                                                                                            | CO-SUPERVISOR'S SIGNATURE |
|                                      |            | Signature & Date                                                                                                                                                  | Signature & Date          |

| APRIL 2026                           |            | 10 ENTRIES                                                                                                                                                               |                           |
|--------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| NO.                                  | DATE       | RP WORK CARRIED OUT                                                                                                                                                      |                           |
| 6                                    | 02.04.2026 | Designed fallback behaviour so the developer agent could still produce usable outputs when the preferred model path is unavailable.                                      |                           |
| 7                                    | 04.04.2026 | Extended the design to include prompt-driven generation for both components and pages using structured SRS context.                                                      |                           |
| 8                                    | 07.04.2026 | Refined the repair strategy to focus on controlled bug fixing and stable re-generation instead of broad rewrites.                                                        |                           |
| 9                                    | 09.04.2026 | Planned runtime sandbox support so generated React applications could be previewed, refined and repaired through an isolated execution layer.                            |                           |
| 10                                   | 11.04.2026 | Coordinated with the group on the shared handoff from requirements analysis and design selection into Builder Studio.                                                    |                           |
| 11                                   | 25.04.2026 | Implemented the Agent 2 backend in code, including build start/status/file endpoints, background run tracking, output packaging and the LangGraph-based generation flow. |                           |
| 12                                   | 25.04.2026 | Integrated component and page generation prompts, style variants, catalogue endpoints and manifest packaging for produced artefacts.                                     |                           |
| 13                                   | 26.04.2026 | Implemented sandbox planning, generation, refine and repair support with a preloaded UI library and HTML preview wrapper for frontend runtime validation.                |                           |
| 14                                   | 27.04.2026 | Supported the shared group demo by aligning Builder Studio outputs with the common dashboard, activity feed and inter-agent navigation.                                  |                           |
| 15                                   | 28.04.2026 | Consolidated Agent 2 outcomes for the period and reviewed how the builder workflow fits into the first completion milestone of the project.                              |                           |
| MONTHLY VERIFICATION — APRIL 2026    |            |                                                                                                                                                                          |                           |
| SUPERVISOR'S COMMENTS FOR APRIL 2026 |            | SUPERVISOR'S SIGNATURE                                                                                                                                                   | CO-SUPERVISOR'S SIGNATURE |
|                                      |            | Signature & Date                                                                                                                                                         | Signature & Date          |

| MAY 2026                           |            | 6 ENTRIES                                                                                    |
|------------------------------------|------------|----------------------------------------------------------------------------------------------|
| NO.                                | DATE       | RP WORK CARRIED OUT                                                                          |
| 16                                 | 04.05.2026 | Set up the builder front-end base routes and the Studio dashboard shell.                     |
| 17                                 | 04.05.2026 | Reorganised the front-end folder structure and removed the superseded starter application.   |
| 18                                 | 05.05.2026 | Implemented the dashboard cards, page flow and overall layout for the shared workspace.      |
| 19                                 | 05.05.2026 | Added the QA cycle timeline and testing progress states to the shared dashboard views.       |
| 20                                 | 09.05.2026 | Built the shared builder components and wired the builder routes into the common navigation. |
| 21                                 | 10.05.2026 | Completed responsive checks and visual polish across the builder screens.                    |
| MONTHLY VERIFICATION — MAY 2026    |            |                                                                                              |
| SUPERVISOR'S COMMENTS FOR MAY 2026 |            | SUPERVISOR'S SIGNATURE                                                                       |
|                                    |            | Signature & Date                                                                             |
|                                    |            | CO-SUPERVISOR'S SIGNATURE                                                                    |
|                                    |            | Signature & Date                                                                             |

| JUNE 2026                           |            | 13 ENTRIES                                                                                                                                                                                                                |                           |
|-------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| NO.                                 | DATE       | RP WORK CARRIED OUT                                                                                                                                                                                                       |                           |
| 22                                  | 01.06.2026 | Added the AI developer workflow and the architect agent that turns approved requirements into an implementation plan.                                                                                                     |                           |
| 23                                  | 01.06.2026 | Introduced generation state management and added the required-field rules to the architect prompt.                                                                                                                        |                           |
| 24                                  | 02.06.2026 | Added multi-stage architect and planner agents so high-level design is separated from file-level planning.                                                                                                                |                           |
| 25                                  | 14.06.2026 | Replaced the first developer-agent prototype with the ai_designer scaffold as the version 2 baseline.                                                                                                                     |                           |
| 26                                  | 16.06.2026 | Established the architecture stability baseline: error-text decoding, route resolver, synchronous endpoints, element-scoped LLM patching with self-heal, a 102-entry template catalogue and version-on-success behaviour. |                           |
| 27                                  | 16.06.2026 | Reskinned the Studio interface on a Tailwind v4 foundation with neutral-dark theme tokens.                                                                                                                                |                           |
| 28                                  | 16.06.2026 | Centralised and unit-tested error handling so runtime failures never surface as unreadable objects.                                                                                                                       |                           |
| 29                                  | 16.06.2026 | Implemented a cache-busted image and build flow and consolidated the stability test suite.                                                                                                                                |                           |
| 30                                  | 17.06.2026 | Added the component registry, versioning and edit-plan support needed for controlled regeneration.                                                                                                                        |                           |
| 31                                  | 17.06.2026 | Hardened component selection with a testable route-fallback library and a CRUD create resolver.                                                                                                                           |                           |
| 32                                  | 18.06.2026 | Added backend regression tests and scaffold updates for the generation pipeline.                                                                                                                                          |                           |
| 33                                  | 23.06.2026 | Merged the stability fix branch, removed dead code and added further project templates.                                                                                                                                   |                           |
| 34                                  | 28.06.2026 | Divided the agent implementation into task-specific modules to improve maintainability.                                                                                                                                   |                           |
| MONTHLY VERIFICATION — JUNE 2026    |            |                                                                                                                                                                                                                           |                           |
| SUPERVISOR'S COMMENTS FOR JUNE 2026 |            | SUPERVISOR'S SIGNATURE                                                                                                                                                                                                    | CO-SUPERVISOR'S SIGNATURE |
|                                     |            | Signature & Date                                                                                                                                                                                                          | Signature & Date          |

| JULY 2026 |            | 4 ENTRIES                                                                                                         |
|-----------|------------|-------------------------------------------------------------------------------------------------------------------|
| NO.       | DATE       | RP WORK CARRIED OUT                                                                                               |
| 35        | 07.07.2026 | Reorganised the application workspace and tightened the tracked-file rules for the desktop build.                 |
| 36        | 09.07.2026 | Upgraded the planner so implementation plans cover file layout, dependencies and build order more completely.     |
| 37        | 17.07.2026 | Changed the model configuration used by the developer agent after comparing generation quality across candidates. |
| 38        | 18.07.2026 | Implemented architect-driven planning with split model routing so each stage uses the most suitable model.        |

| MONTHLY VERIFICATION — JULY 2026    |                        |                           |
|-------------------------------------|------------------------|---------------------------|
| SUPERVISOR'S COMMENTS FOR JULY 2026 | SUPERVISOR'S SIGNATURE | CO-SUPERVISOR'S SIGNATURE |
|                                     | Signature & Date       | Signature & Date          |

| AUGUST 2026 |            | 11 ENTRIES                                                                                                                                |
|-------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| NO.         | DATE       | RP WORK CARRIED OUT                                                                                                                       |
| 39          | 14.08.2026 | Integrated the agent modules into a single package and removed the obsolete test scaffolding.                                             |
| 40          | 16.08.2026 | Recorded a lessons store and added a browser reproducer for defects observed in generated applications.                                   |
| 41          | 16.08.2026 | Tuned the generation prompts, updated the Studio interface and added the console bug-fixer flow.                                          |
| 42          | 20.08.2026 | Added workspace tools and source-guidance support so the agent reuses existing project context when editing.                              |
| 43          | 21.08.2026 | Added the theme, photo and image agent modules for asset-aware generation.                                                                |
| 44          | 21.08.2026 | Split the agent exports into focused submodules to reduce coupling across the pipeline.                                                   |
| 45          | 22.08.2026 | Added persistence and builder scaffolding for project state.                                                                              |
| 46          | 22.08.2026 | Split the tester modules and added a security shim for the QA-facing interface.                                                           |
| 47          | 22.08.2026 | Merged the developer-agent parts and the full-application developer edition into the shared integration branch.                           |
| 48          | 23.08.2026 | Performed history-only merges to restore the QA and SRS component branches into the integration line.                                     |
| 49          | 25.08.2026 | Reviewed the integrated Studio interface across the builder, SRS, QA and deployment panes in preparation for the version 2 consolidation. |

| MONTHLY VERIFICATION — AUGUST 2026    |                        |                           |
|---------------------------------------|------------------------|---------------------------|
| SUPERVISOR'S COMMENTS FOR AUGUST 2026 | SUPERVISOR'S SIGNATURE | CO-SUPERVISOR'S SIGNATURE |
|                                       | Signature & Date       | Signature & Date          |

#### 4. CONSOLIDATED MONTHLY VERIFICATION

| MONTH        | ENTRIES<br>LOGGED | SUPERVISOR'S SIGNATURE | DATE |
|--------------|-------------------|------------------------|------|
| March 2026   | 5                 |                        |      |
| April 2026   | 10                |                        |      |
| May 2026     | 6                 |                        |      |
| June 2026    | 13                |                        |      |
| July 2026    | 4                 |                        |      |
| August 2026  | 11                |                        |      |
| <b>Total</b> | <b>49</b>         |                        |      |

#### 5. PROJECT WORK PLANNED BEFORE THE NEXT EVALUATION PERIOD

1. Complete the version 2 integration so all four components run behind the single Studio interface and shared runtime without branch-specific behaviour.
2. Improve sandbox runtime debugging, preview reliability and automated repair quality for generated applications.
3. Extend the generated output to a fuller set of backend and frontend artefacts driven directly from the approved SRS and plan.
4. Strengthen reuse so the agent learns from stored lessons and reference projects and transfers those patterns systematically.
5. Raise pre-handoff validation so builds reach Agent 3 with fewer structural and runtime defects, and finish the console bug-fixer loop.

#### 6. DECLARATION AND SIGNATURES

I certify that the work recorded in this log book for the period 23 March 2026 to 25 August 2026 was carried out by me as the individual component contribution to research project R26-SE-009, and that the group-level contributions recorded here were made jointly with the other project members.

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|-----------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>STUDENT</b><br><hr/> Ravindu Bandara Subasinha (IT22098450) — Signature & Date | <b>SUPERVISOR</b><br><hr/> Prof. Nuwan Kodagoda — Signature & Date |
| <b>CO-SUPERVISOR</b><br><hr/> Mr. Eishan Weerasinghe — Signature & Date           | <b>OVERALL SUPERVISOR COMMENTS</b><br><hr/> Comments               |